

HAJIR MOHAMMED

Control Systems · UAV Guidance, Navigation & Control

hajermohamed505@gmail.com | +90 538 079 13 33 | Kayaşehir, Istanbul, Türkiye

PROFILE

Final-year Control Engineering student at Yıldız Technical University with direct, hands-on experience in autonomous UAV systems — spanning VTOL airframe design, flight control algorithm development, sensor fusion, and embedded implementation on STM32 platforms. Proficient in MATLAB/Simulink-based model-driven development, controller design and tuning, and system identification, with practical exposure to flight controllers (SpeedyBee), open-source autopilot stacks (iNAV, Mission Planner), and C/Embedded C. Driven to contribute to the design and validation of control, guidance, and navigation algorithms for fixed-wing UAV platforms in a high-performance R&D environment.

EDUCATION

Yıldız Technical University — Istanbul, Türkiye

Expected December 2026

Bachelor of Control Engineering

Senior Thesis: Design and Control of a VTOL (Vertical Take-Off and Landing) Unmanned Aerial Vehicle

Relevant Coursework: Linear Control Systems, System Dynamics, Feedback Control Theory, Digital Control, Flight Mechanics, Signal Processing, Embedded Systems, Modeling & Simulation, Mechatronic Systems.

PROJECTS & ENGINEERING EXPERIENCE ([LINK](#))

VTOL UAV — Design, Modeling & Flight Control . Senior Thesis Project

Oct 2024 – Jun 2025

- Designed and 3D-printed a custom VTOL airframe and lifting surfaces; integrated propulsion, avionics, and flight controller for full vertical-to-forward flight transition.
- Developed a full nonlinear 6-DOF dynamic model of the airframe in MATLAB/Simulink, including aerodynamic, propulsion, and actuator subsystems for high-fidelity simulation.
- Performed system identification on motor and airframe response data to derive validated transfer-function and state-space models used for controller synthesis.
- Designed and tuned cascaded attitude, altitude, and position controllers (PID + state-space); verified stability margins through Bode, Nyquist, and root-locus analysis.
- Implemented a Kalman / complementary filter sensor-fusion algorithm to estimate attitude and orientation from raw IMU data.
- Deployed control logic on an STM32 microcontroller (STM32CubeIDE); configured and tuned a SpeedyBee flight controller running iNAV, with mission and PID tuning through Mission Planner.

MATLAB Mini Drone Competition · MathWorks ([LINK](#))

May 2024

- Designed and simulated quadrotor flight-control algorithms in MATLAB/Simulink for the official MathWorks competition framework.
- Developed trajectory-tracking and waypoint-navigation logic; iteratively tuned controllers to improve settling time, overshoot, and disturbance rejection.
- Collaborated within a multidisciplinary team to benchmark autonomous flight performance against competition success metrics.

- DC motor speed/position control: modeled the plant in Simulink and tuned closed-loop controllers for accurate reference tracking.
- Ball-and-beam balancing: designed and optimized a PID controller for a nonlinear, underactuated system.
- Magnetic levitation: derived a linearized state-space model and implemented a stabilizing controller for an inherently unstable plant.
- Coupled-tank system: implemented multi-loop liquid-level control and validated step response and disturbance rejection experimentally.

Digital Oscilloscope Modeling Project · Coursework

2024

- Designed and simulated a digital oscilloscope model in MATLAB for signal acquisition and visualization.
- Tuned sampling, triggering, and signal-processing parameters to optimize fidelity of captured waveforms.

TECHNICAL SKILLS

Simulation & Modeling	MATLAB, Simulink, Simscape, Control System Toolbox, System Identification Toolbox
Control Theory	Linear Control Systems, PID, State-Space, Pole Placement, Cascaded Control, Frequency-Domain Analysis (Bode, Nyquist, Root Locus), Digital Control, System Identification
Estimation & Sensor Fusion	Kalman Filter, Complementary Filter, IMU-based Attitude Estimation
Programming	C, Embedded C, MATLAB scripting, basic Python
Embedded & Hardware	STM32 (STM32CubeIDE), Arduino, SpeedyBee Flight Controller, IMU sensors, UART / I ² C / SPI
UAV Toolchain	iNAV, Mission Planner, Ardupilot ecosystem, 3D printing for airframe fabrication
Engineering Workflow	Model-Based Design, Software-in-the-Loop testing, technical documentation & reporting

LANGUAGES

Arabic (Native) English (Fluent) Turkish (Fluent)

AREAS OF INTEREST

Autonomous flight systems · Fixed-wing & VTOL UAV control · Guidance and navigation algorithms · Sensor fusion and state estimation · Flight simulation and model-based design